

Telecom Churn Intelligence Dashboard

Analytics n = 7,043 customers IBM Telco Dataset

7,043

Total Customers

26.54%

Churn Rate

73.46%

Retention Rate

64.76

Avg Monthly Charges

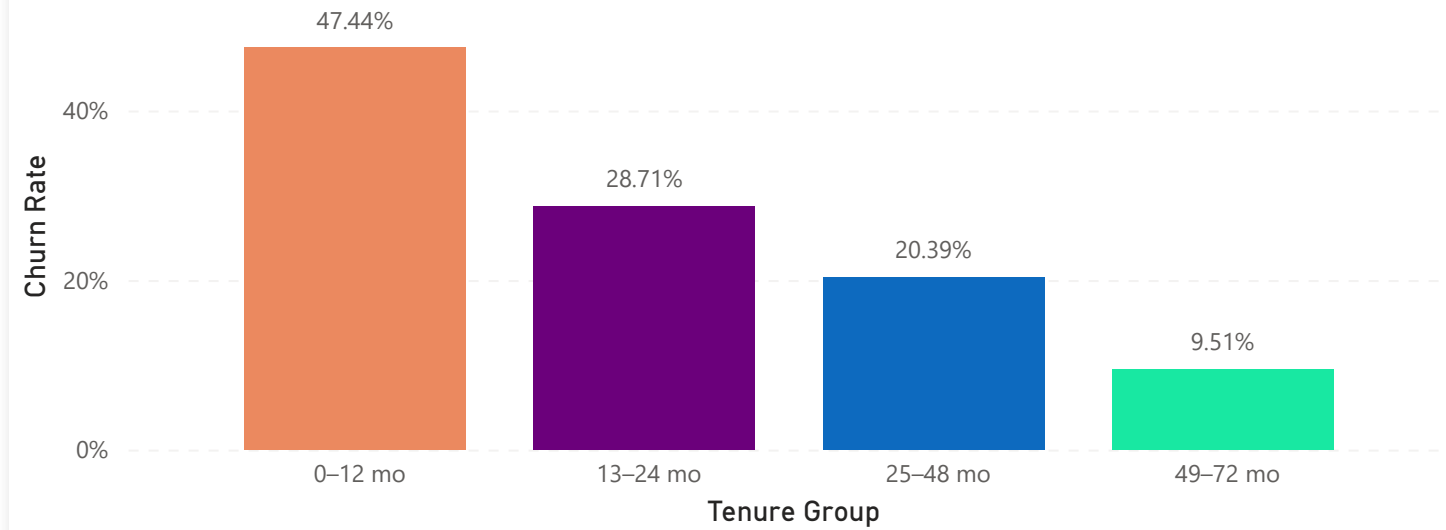
\$139.13K

Revenue at Risk

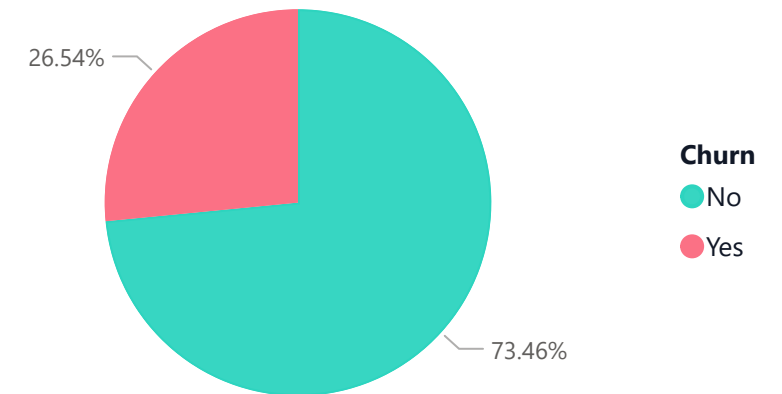
37.57

Avg Tenure (Retained)

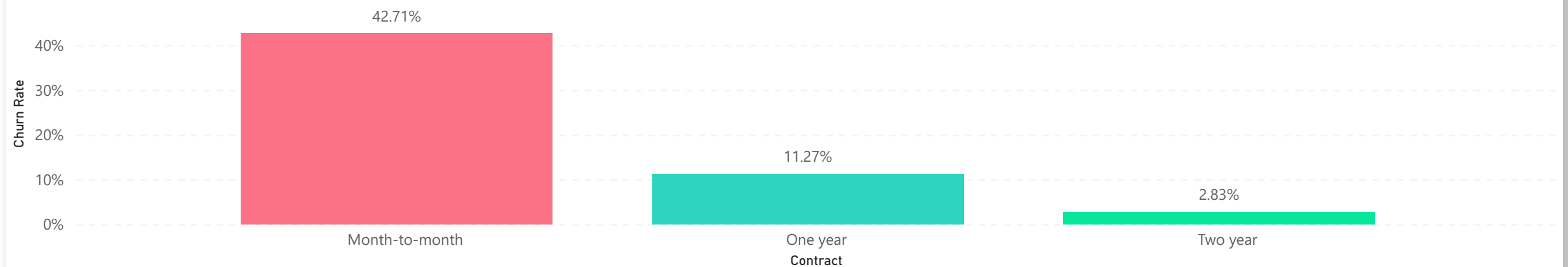
Churn by Tenure Group



Churn vs Retained



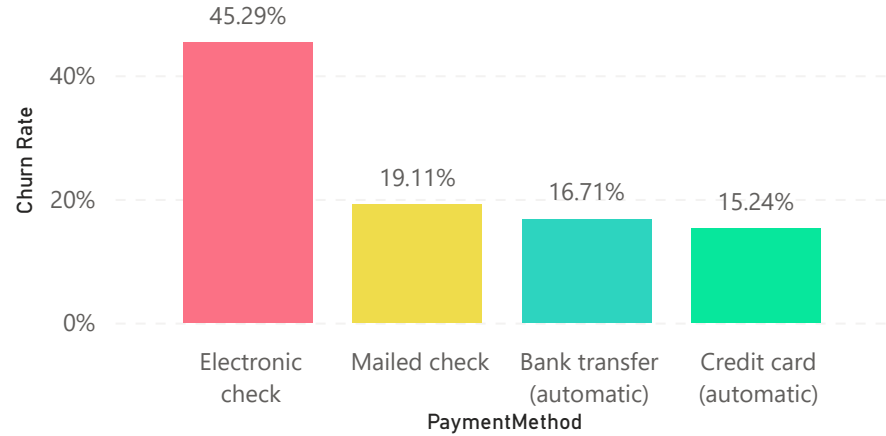
Churn Rate by Contract



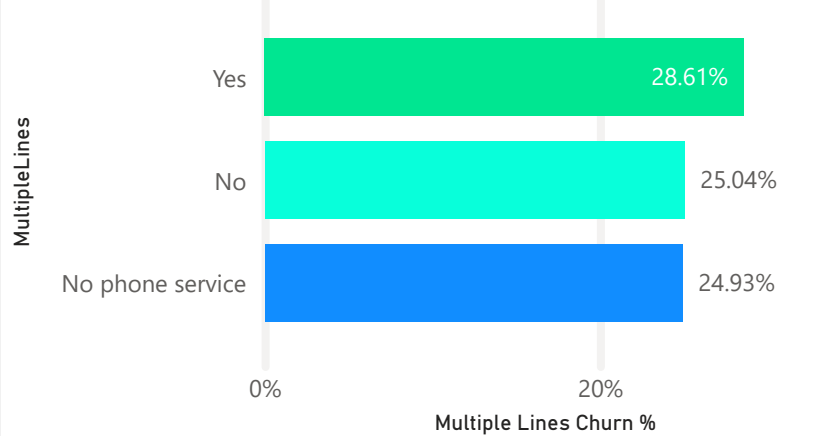
Churn Factor Analysis

Identifying key drivers of customer attrition across all dimensions

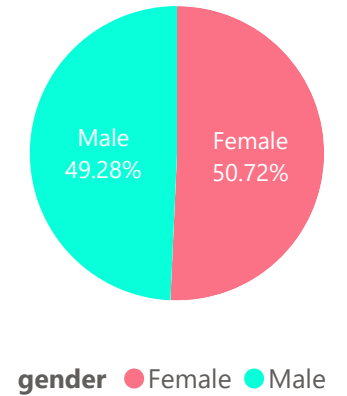
Churn Rate by PaymentMethod



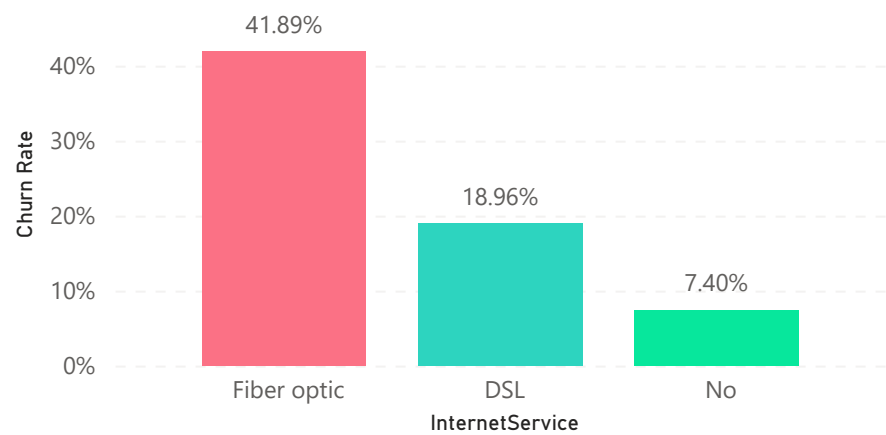
Multiple Lines Service vs Churn



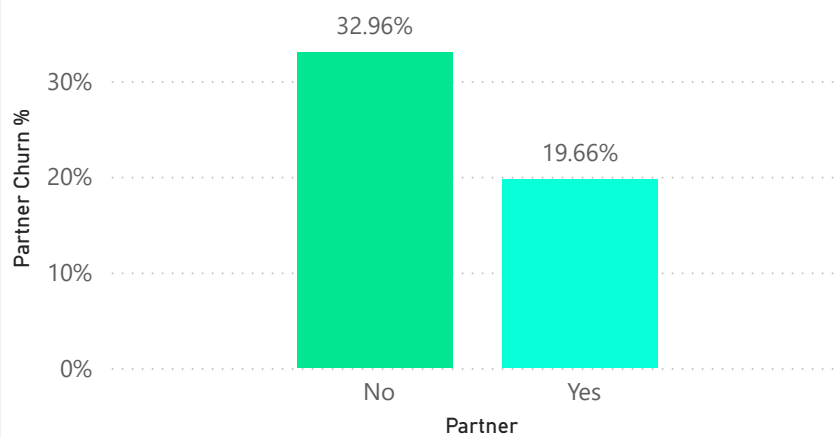
Gender-wise Churn Rate



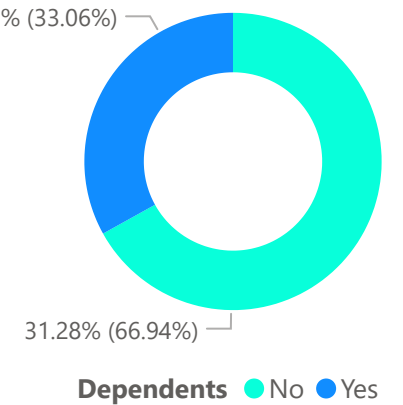
Churn Rate by InternetService



Partner Status vs Churn



Dependents vs Churn



Revenue Analysis

Contract

All

PaymentMethod

All

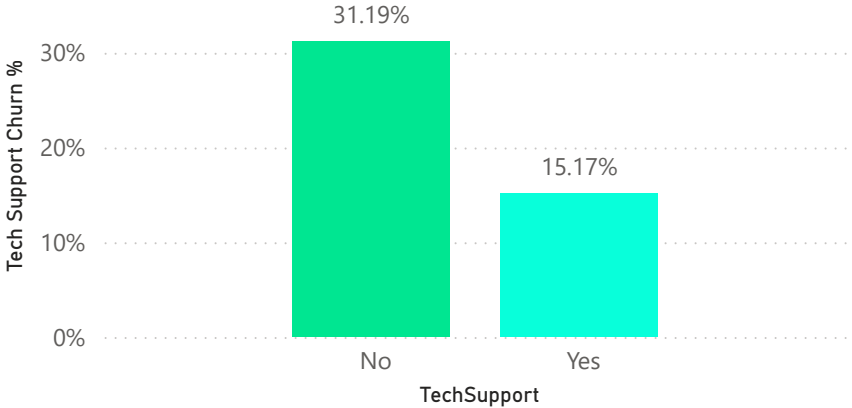
InternetService

All

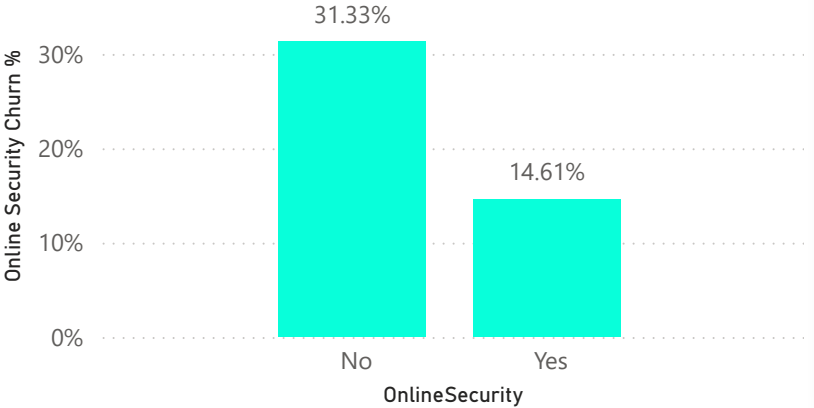
gender

All

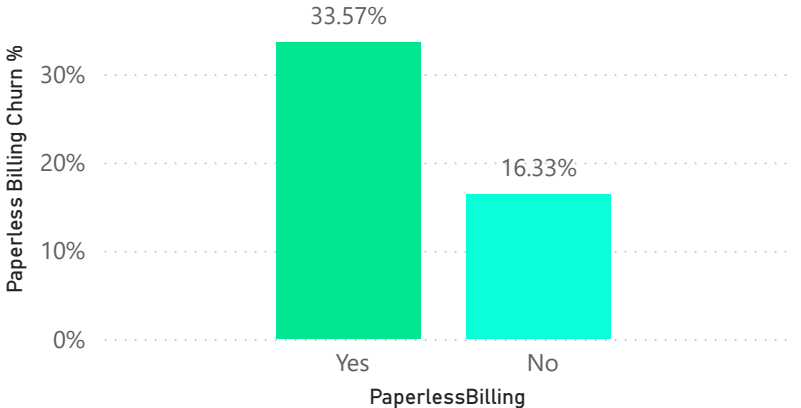
Tech Support Impact on Churn



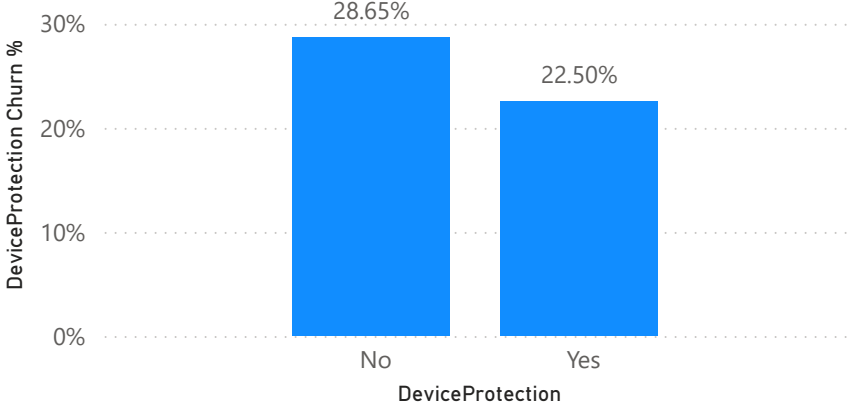
Online Security vs Churn



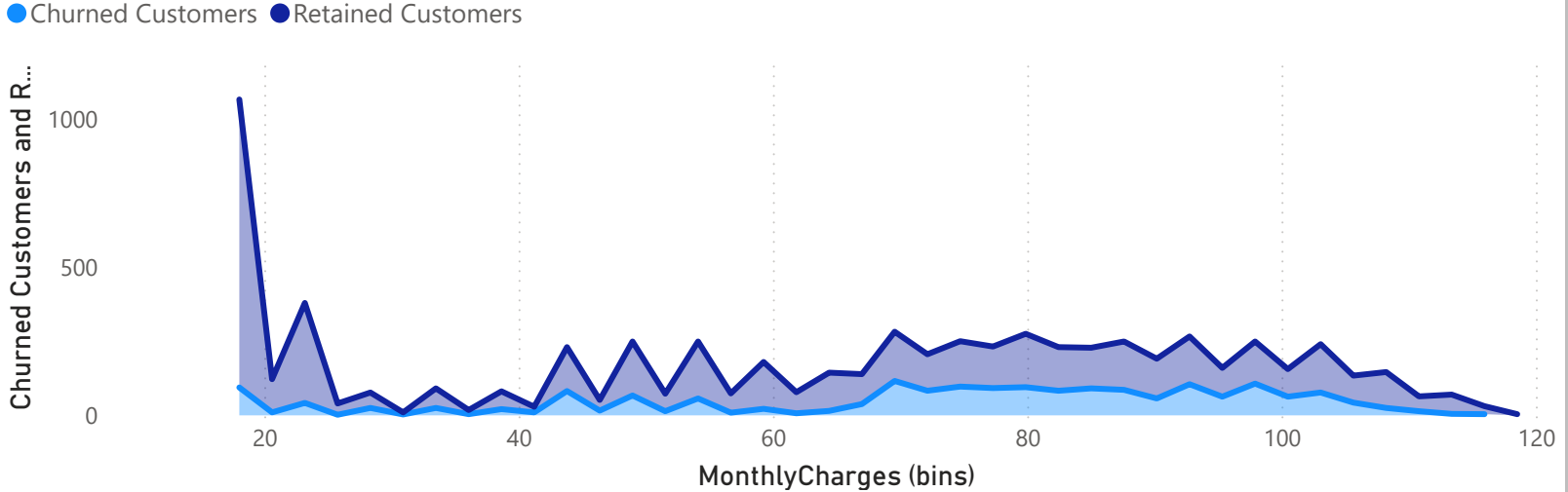
Paperless Billing vs Churn



Device Protection Impact on Churn

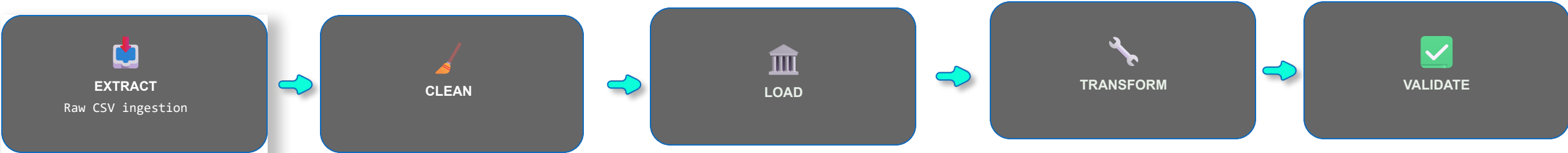


Monthly Charges Distribution — Churned vs Retained



ETL Pipeline and Data Warehouse

Extract · Transform · Load · Validate · Star schema architecture ETL Pipeline Architecture



Data Quality Metrics (Post-ETL)

Check	Before ETL	After ETL	Status
Churn Encoding	Yes/No	Binary 0/1	PASS
Duplicate Rows	0	0	PASS
Null Values	11 (TotalCharges)	0	PASS
Row Count Preserved	7043	7043	PASS
Tenure Bucketed	No	4 Cohorts	PASS
TotalCharges Type	String	Float64	PASS

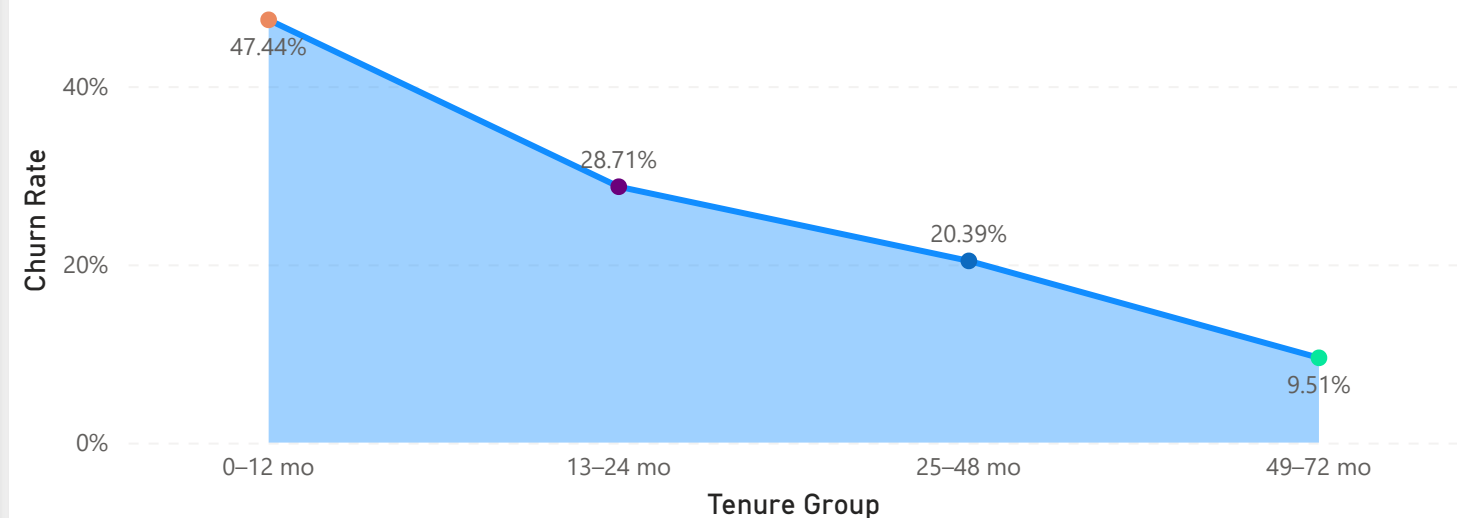
Customer Segmentation

High-risk cohorts identified across contract type, payment, demographics and services

Segment Risk Matrix

Segment	Sum of Churn Rate	Sum of Customers	First Risk Level
Total	324.69%	26521	CRITICAL
+ 0–12 Months Tenure	47.68%	2175	CRITICAL
+ Electronic Check	45.29%	2365	CRITICAL
+ Month-to-Month	42.71%	3875	CRITICAL
+ Fiber Optic	41.89%	3096	CRITICAL
+ No Online Security	41.77%	3500	HIGH
+ Senior Citizens	41.68%	1142	CRITICAL
+ Paperless Billing	33.57%	4200	HIGH
+ Auto-Pay Users	16.00%	3000	SAFE
+ One-Year Contract	11.27%	1473	LOW
+ Two-Year Contract	2.83%	1695	SAFE

Churn by Tenure Group



Build visuals with your data

Select or drag fields from the **Data** pane onto the report canvas.

